Friedrich-Alexander-Universität
Technische Fakultät

**Department of Computer Science**
Computer Science 6
(Data Management)

**Knowledge Discovery in Databases with Exercises**
**Summer Semester 2025**

# Exercise Sheet 4: Classification

## About this Exercise Sheet

This exercise sheet focuses on the content of lecture *7. Classification*.

It includes both theoretical exercises on Decision Trees (Exercise 1) and Naïve Bayes (Exercise 2) and a practical data science exercise (Exercise 3).

The exercise sheet is designed for a three-week period, during which the tasks can be completed flexibly (Planned is one exercise per week).

The sample solution will be published after the three weeks have elapsed.

## Preparation

Before participating in the exercise, you must prepare the following:

1. **Install Python and pip on your computer**
   - Detailed instructions can be found in `1-Introduction-Python-Pandas.pdf`.
2. **Download provided additional files**
   - Download `Additional-Files-Student.zip` from StudOn
   - Extract it to a folder of your choice.
3. **Install required Python packages**
   - Open a terminal and navigate to the folder where you extracted the files.
   - Run the command `pip install -r requirements.txt` within the extracted additional files folder to install the required Python packages.

# Exercise 1: Decision Trees

Given is a dataset $D$.

$D$ is containing a continuous attribute (*Age*) and two categorical attributes (*Major* and *Participation*) which can be used to predict the target attribute *Passed*.

| Age | Major | Participation | Passed |
|---|---|---|---|
| 23 | CS | High | Yes |
| 23 | DS | Low | No |
| 26 | DS | High | Yes |
| 24 | DS | Medium | Yes |
| 26 | DS | Medium | No |
| 26 | DS | Low | No |

## Task 1: Information Gain

Use the algorithm for **Decision Tree Induction** known from the lecture to build a decision tree for dataset $D$. The decision tree should be built using **Information Gain** as the attribute selection method.

Write down **all** intermediate steps.

## Task 2: Gini Index

This time, the decision tree for dataset $D$ should be built using the **Gini Index** as the attribute selection method.

### Task 2.a: Root Node

Using the algorithm for building a decision tree with the Gini Index, create the root node of the decision tree for the dataset $D$.

Write down **all** intermediate steps **up to** (and including) the point where the root node is created.

### Task 2.b: Splitting attribute candidates

In the resulting tree from Task 2.a, one of the branches is already a leaf node.

Which of the **attributes** *Age*, *Major* and *Participation* have to **be checked** for their Gini index in the next step necessary to further split the remaining branch?

## Task 3: Gain Ratio

The **Gain Ratio** is a solution to a problem of the **Information Gain**.

Come up with an example **dataset** showing the problem of the Information Gain and explain how the Gain Ratio solves this problem.

## Exercise 2: Naïve Bayes

Given is a dataset $D$.

It can be assumed that *Topic*, *Knowledge* and *Hours* are conditionally independent of each other.

The attributes *Topic* and *Knowledge* are categorical attributes.

The attribute *Hours* is a continuous attribute. It can be assumed that the values of this attribute are distributed according to a Gaussian distribution.

| Topic | Knowledge | Hours | Passed |
|---|---|---|---|
| Classification | High | 1,0 | No |
| Clustering | Low | 4,0 | No |
| Frequent Patterns | High | 5,0 | Yes |
| Clustering | Medium | 5,0 | Yes |
| Frequent Patterns | High | 2,0 | No |
| Frequent Patterns | Medium | 3,0 | Yes |
| Classification | Low | 6,0 | Yes |
| Clustering | Low | 5,0 | Yes |
| Clustering | High | 3,0 | Yes |
| Classification | Medium | 4,0 | Yes |

### Task 1: Classification

Use the dataset $D$ and the Naïve Bayes algorithm to classify the following tuples:

| Topic | Knowledge | Hours | Passed |
|---|---|---|---|
| Clustering | Medium | 4,0 | ? |
| Classification | High | 3,0 | ? |
| Frequent Patterns | Low | 6,8 | ? |

Write down **all** intermediate steps.

### Task 2: Model Evaluation

The classifier was also trained on a version of dataset $D$ with more tuples:

The dataset $T$ contains both the true and the predicted "Passed"-Status for each test tuple.

| Topic | Knowledge | Hours | Passed (True) | Passed (Pred) |
|---|---|---|---|---|
| Classification | Medium | 7,5 | Yes | Yes |
| Frequent Patterns | Low | 1,8 | No | No |
| Frequent Patterns | High | 3,7 | No | Yes |
| Frequent Patterns | Low | 0,2 | No | No |
| Frequent Patterns | High | 1,4 | Yes | No |
| Frequent Patterns | High | 9,9 | Yes | Yes |
| Frequent Patterns | Medium | 7,3 | Yes | Yes |
| Frequent Patterns | Low | 4,3 | No | Yes |
| Classification | Medium | 5,5 | Yes | Yes |
| Clustering | Low | 0,1 | No | No |

Use the dataset $T$ to calculate the **sensitivity**, **specificity**, **accuracy**, **precision**, **recall**, and **F1-score** of the model.

Also state the **best possible** value for each metric.

## Exercise 3: Conducting Classification

This exercise comprises practical data science tasks and thus utilizes a Jupyter Notebook:

1. Open `Conducting-Classification.ipynb`.
2. Take a look at the tasks (blue boxes) in the notebook and try to solve them.

If you are unfamiliar with how to open a Jupyter Notebook, please refer to Exercise 1 of `1-Introduction-Python-Pandas.pdf`.